

Basic research questions:.....

At what extend is chess playing possible under the Machine Coaching paradigm and how generalizable are the results acquired? Is it possible to generalize this approach to more complex (in terms of rules or game search/state space or other complexity metrics) games? What are the boundaries of the Machine Coaching paradigm in the domain of game playing (empirically and/or theoretically evaluated)?

Necessity and importance of research (Αναγκαιότητα και σπουδαιότητα της έρευνας):.....

The necessity and importance of the proposed research is evident under the light of the vast need for a more explainable AI and, hence, a deeper understanding of the way reasoning takes place in autonomous/intelligent systems. Moreover, the Machine Coaching paradigm enables the active participation of the (human) instructor in the learning procedure of the machine (Human-Computer interaction), hence, enhancing not only the understanding of the reasoning process of the machine but also the efficiency of the learning procedure, by letting the machine be explicitly “aware” of the reasons behind any unwanted behavior it adopts. Also, under this paradigm, the human coach is given the possibility to explicitly counter-argue against the arguments presented by the machine, contributing towards a more explainable behavior by the machine itself.

Suggested methodology: The research will begin with an implementation of the typical Machine Coaching cycle in Java and will continue with the development of a domain specific chess playing platform under that paradigm. The platform will be a single player platform, where one trains the machine to play against other players (human or not). At a second stage of development (see timetable, below), it is possible that the platform is further enhanced so as to support a multi-player bot-versus-bot functionality.

Timetable: Today – 23/10/2019: Transcription of the prolog based Machine Coaching cycle into Java
24/10/2019 – 28/11/2019: Initial development of the chess playing platform.

29/11/2019 – 31/12/2019: First evaluation of the platform – experiments with human users.

01/01/2020 – end of academic year: Further development of the platform/further evaluation.

Thematic Unit/Thematic Field: Machine Coaching.

Date: 16/11/2019.....

Student's Signature: .....